

INTERNSHIP REPORT

Medical Appointment No-Show Prediction Using Machine Learning

Exploratory Data Analysis • Temporal Preprocessing • Comparative Modeling
Decision Tree • Random Forest • Gradient Boosting

Host Company	AnalystLAB
Objective	Binary classification of appointments: Attended vs No-Show
Dataset	5,000 medical appointments (synthetic data)
Approach	Strict temporal split + tree-based models (no hyperparameter tuning)
Best Model	Decision Tree (Accuracy 61.1% • F1-Score 0.555)

Prepared during the internship at AnalystLAB
September 2026

1. Introduction and Objectives

Unjustified absences (No-Shows) at medical appointments represent a major challenge for healthcare facilities: loss of productivity, longer waiting lists, and under-utilization of resources. The objective of this project, carried out during the internship at **AnalystLAB**, is to build a predictive model capable of anticipating, from information available at the time of booking, whether a patient will attend the appointment or not.

The work was structured into three main phases:

- **Exploratory Data Analysis (EDA)**: understanding data structure, detecting anomalies and identifying predictive signals.
- **Preprocessing**: cleaning, encoding, scaling and, most importantly, a **strict temporal split** to avoid any future information leakage.
- **Comparative modeling**: training of three tree-based algorithms (Decision Tree, Random Forest, Gradient Boosting) with default hyperparameters, followed by rigorous evaluation.

This report presents the entire pipeline in detail and concludes on the best performing model **without any hyperparameter optimization**.

2. Dataset Description

The dataset contains **5,000 appointments** and 18 initial variables. The main features are:

Variable	Type	Description
appointment_outcome	Target	No-Show / Attended / Cancelled → binarized (1 = Attended)
booking_lead_days	Numeric	Number of days between booking and appointment
age / age_group	Num. / Cat.	Patient age and age group
distance_to_clinic_km	Numeric	Distance from home to clinic
previous_no_shows	Numeric	Patient's historical no-show count
previous_appointments	Numeric	Number of previous appointments
waiting_time_minutes	Numeric	Estimated waiting time
reminder_sent / channel	Categorical	Reminder sent and channel (SMS, WhatsApp, Email)
appointment_type / day / time	Categorical	Consultation type, day of week and time slot

Table 1 – Main variables of the dataset

The target was binarized: the modalities “No-Show” and “Cancelled” were grouped together against “Attended”. This simplification is justified by the low proportion of Cancelled cases (~6%) and by the operational objective (presence vs non-presence).

3. Exploratory Data Analysis (EDA)

3.1 Target distribution

The target variable shows a nearly balanced distribution between No-Show (~48%) and Attended (~46%), with a minority of Cancelled cases (~6%). This relative balance facilitates learning without the immediate need for aggressive rebalancing techniques.

3.2 Anomalies and data quality

A notable anomaly was detected: appointments are scheduled on **Sundays**, while the clinic is theoretically closed on that day. This confirms the synthetic nature of the dataset and the presence of imperfect generation rules. Numerical variables show realistic distributions, with a few outliers on distance and historical counts.

3.3 Correlations and non-linearity

The correlation matrix reveals an almost complete absence of strong linear relationships between predictors and the target (except for the logical link `previous_appointments` ↔ `previous_no_shows`). Pairplots confirm high dispersion without clear linear boundaries. **Methodological conclusion:** linear models (logistic regression) are unsuitable; tree-based algorithms are naturally preferred.

3.4 Key insight: booking_lead_days

Booking lead time (`booking_lead_days`) emerges as the strongest predictive signal. Last-minute bookings (0–7 days) are strongly associated with attendance, while medium-to-long-term bookings (31–90 days) generate a significantly higher No-Show rate. This variable dominates all feature importance rankings of the subsequent models.

4. Preprocessing and Temporal Split

4.1 Strict temporal split

Unlike a classic random split, a **temporal split** was applied to respect the chronological nature of the problem and avoid any future information leakage (data leakage).

1. Strict sorting of the data by **appointment_date**.
2. Cut at 80%: the first 4,000 appointments form the training set, the next 1,000 the test set.

Set	Period	Size
Train	2025-01-01 → 2026-03-13	4,000
Test	2026-03-13 → 2026-06-30	1,000

Table 2 – Temporal split periods

4.2 Transformation pipeline

A `ColumnTransformer` was built with two distinct pipelines:

- **Categorical variables:** imputation (`most_frequent`) + `OneHotEncoder` (`drop='first'`, `handle_unknown='ignore'`).
- **Numerical variables:** imputation (`median`) + `StandardScaler`.

Identifier columns (`appointment_id`, `patient_id`) and raw dates were removed. The final transformer was saved (`joblib`) to ensure reproducibility in production.

After transformation, the training set contains 57 features (one-hot encoding of modalities).

5. Modeling (without optimization)

Three tree-based algorithms were trained with their default hyperparameters (no grid search, no tuning):

- **DecisionTreeClassifier** (single decision tree)
- **RandomForestClassifier** (random forest)
- **GradientBoostingClassifier** (gradient boosting)

These choices are motivated by the EDA: absence of linear relationships and ability of trees to capture non-linear interactions and thresholds (especially on `booking_lead_days`).

6. Comparative Evaluation

6.1 Classification metrics

Model	Accuracy	Precision	Recall	F1-Score
Gradient Boosting	0.605	0.588	0.501	0.541
Random Forest	0.606	0.585	0.527	0.554
Decision Tree	0.611	0.593	0.523	0.555

Table 3 – Performance metrics on the test set (untuned models)

The **Decision Tree** achieves the best Accuracy, Precision and F1 scores. Random Forest is very close. Gradient Boosting shows the lowest Recall.

6.2 Confusion matrices

Confusion matrices (positive class = Attended) allow analysis of Type I and Type II errors.

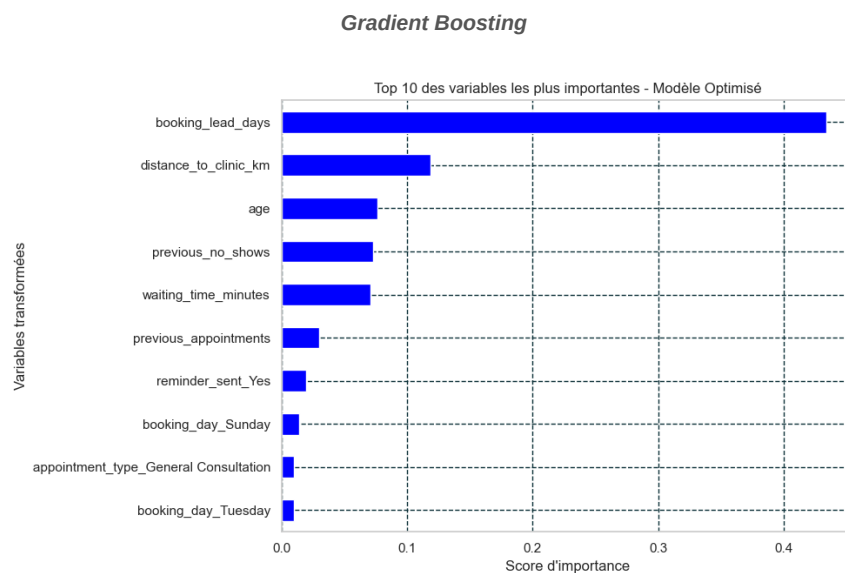


Figure 1 – Confusion matrix – Gradient Boosting (TN=372, FP=163, FN=233, TP=232)

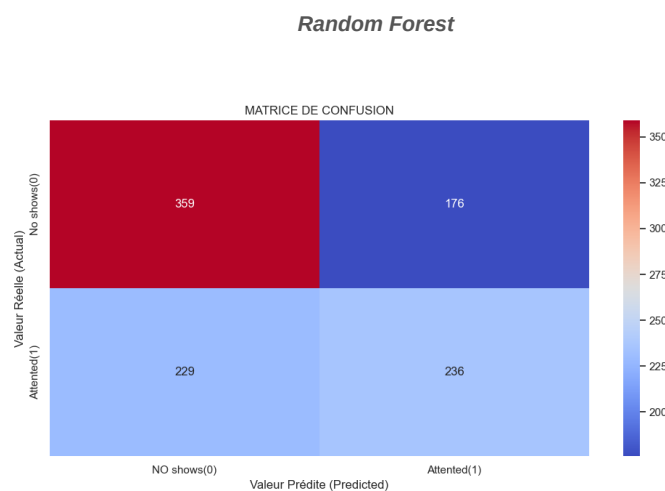


Figure 2 – Confusion matrix – Random Forest (TN=359, FP=176, FN=229, TP=236)

Decision Tree

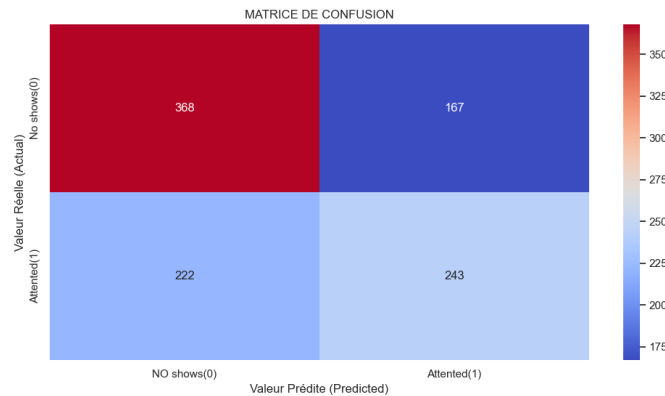


Figure 3 – Confusion matrix – Decision Tree (TN=368, FP=167, FN=222, TP=243)

The Decision Tree maximizes the number of true positives (TP=243) while maintaining a good level of true negatives. It is the most balanced model for detecting attendance.

6.3 ROC and Precision-Recall curves

ROC and PR curves evaluate the models' ability to correctly rank probabilities, independently of the decision threshold.

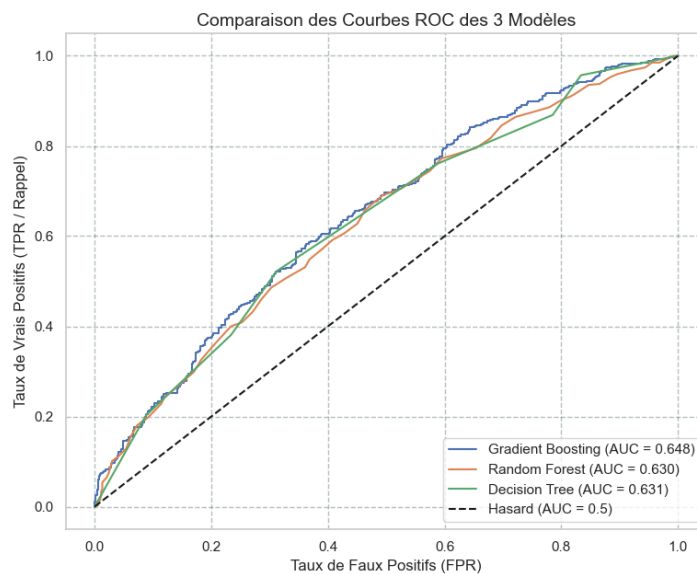


Figure 4 – Comparison of ROC curves (AUC: GB=0.648 • DT=0.631 • RF=0.630)

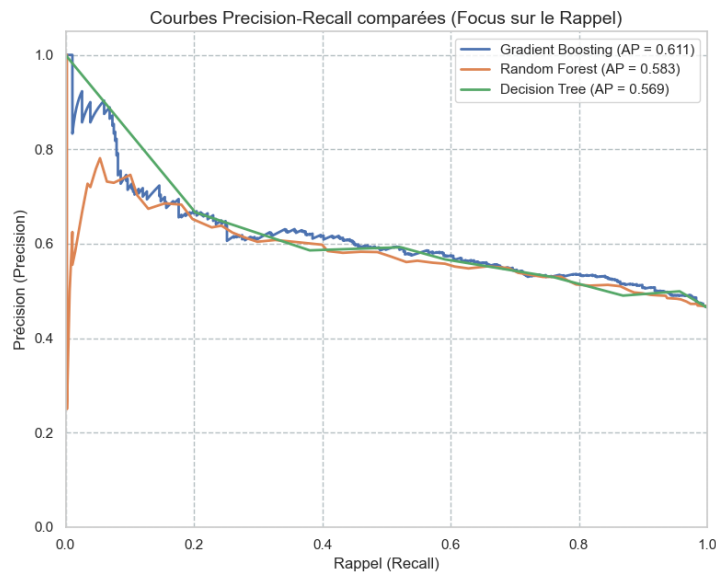


Figure 5 – Precision-Recall curves (Average Precision: GB=0.611 • RF=0.583 • DT=0.569)

On ranking metrics (AUC and Average Precision), **Gradient Boosting** clearly dominates. This indicates better calibration of probability scores. However, for operational use based on clear binary classification (threshold 0.5), Accuracy/F1 metrics remain the most relevant.

7. Feature Importance

Regardless of the importance method (Gini, permutation, SHAP), one variable largely dominates: **booking_lead_days**.

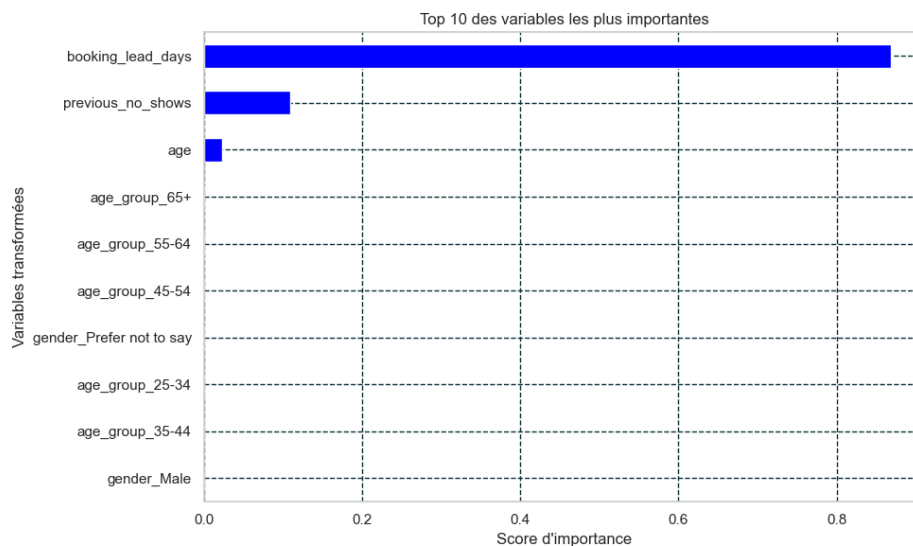


Figure 6 – Top 10 features – Decision Tree

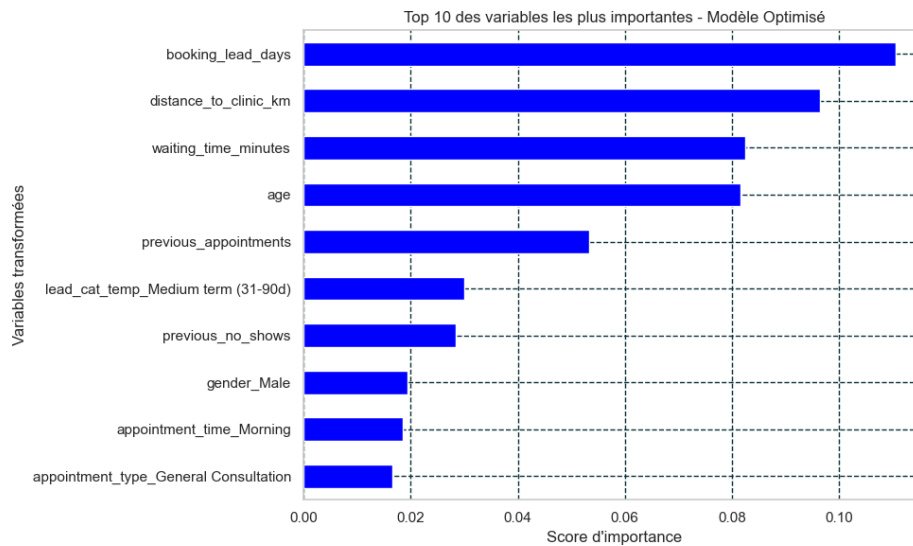


Figure 7 – Top 10 features – Random Forest (optimized model shown for illustration)

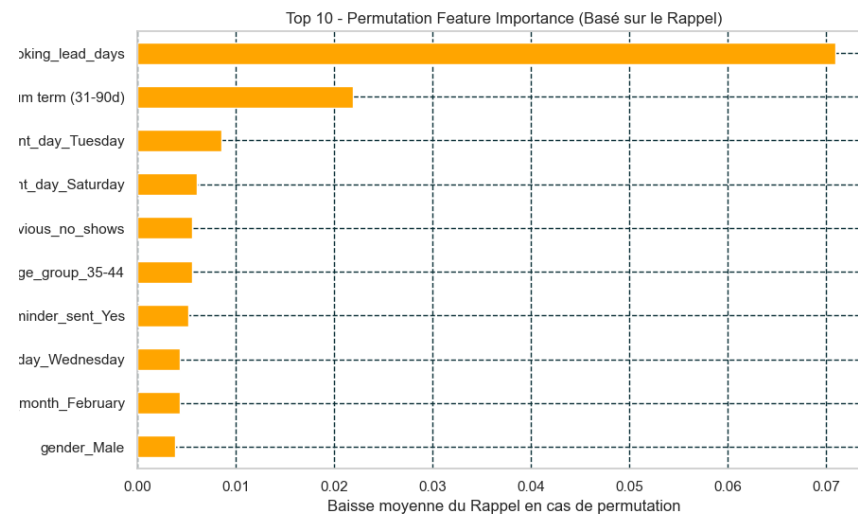


Figure 8 – Permutation Feature Importance (Random Forest) – Focus on Recall

The most recurrent variables in the top 10 are:

1. **booking_lead_days** – booking lead time (dominant signal)
2. **distance_to_clinic_km** – geographic distance
3. **age** and **waiting_time_minutes**
4. **previous_no_shows** and **previous_appointments** – patient history
5. Calendar variables (booking day, consultation type)

These results confirm the EDA insights and validate the relevance of the temporal feature engineering (lead_cat_temp, months, days).

8. Conclusion and Recommendations

8.1 Performance summary (without optimization)

Criterion	Best model
Accuracy / Precision / F1-Score	Decision Tree

Criterion	Best model
Recall	Random Forest / Decision Tree
ROC-AUC & Average Precision	Gradient Boosting
Simplicity & Interpretability	Decision Tree
Overall balance (binary classification)	Decision Tree

Table 4 – Comparative summary

Final recommendation: the Decision Tree is retained as the best model without optimization.

Justification:

- Best classical classification scores (Accuracy 61.1%, F1 0.555).
- Highest number of true positives detected.
- Simplest, fastest to train and most easily interpretable model.
- Gains from Gradient Boosting mainly appear on ranking metrics, which are less critical for operational use with a fixed threshold.

8.2 Limitations and perspectives

Performance remains modest (~61% Accuracy). Several improvement avenues have been identified:

- Hyperparameter optimization (GridSearch / Optuna) – not performed in this version.
- Additional feature engineering (interactions, finer seasonality features).
- Specific treatment of residual imbalance or probability calibration.
- Temporal cross-validation (TimeSeriesSplit) for a more robust estimate of generalization.
- Integration of external data (weather, traffic, public holidays) if available in production.

8.3 Deliverables

The following artifacts were produced and saved:

- Trained models: decision_tree.pkl, random_forest.pkl
- Preprocessing pipeline: column_transformer.pkl
- Transformed datasets (X_train, X_test, y_train, y_test)
- Evaluation plots (confusion matrices, ROC, PR, feature importances)

End of report – Internship at AnalystLAB – September 2026